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Sum of two squares

Q: Which positive integers n can be expressed as

$$n = x^2 + y^2, \quad x, y \in \mathbb{Z} \ ?$$

n	$\square + \square$	n	$\square + \square$
1	$1^2 + 0^2$	13	$2^2 + 3^2$
2	$1^2 + 1^2$	14	X
3	X	15	X
4	$2^2 + 0^2$	16	$4^2 + 0^2$
5	$1^2 + 2^2$	17	$1^2 + 4^2$
6	X	18	$3^2 + 3^2$
7	X	19	X
8	$2^2 + 2^2$	20	$2^2 + 4^2$
9	$3^2 + 0^2$	21	X
10	$1^2 + 3^2$	22	X
11	X	23	X
12	X	24	X

Guess: For an odd prime p, $p = \square + \square$ if and only if $p \equiv 1 \pmod 4$.

Recall that $-1 \,R\, p$ if and only if $p \equiv 1 \pmod 4$.

Maybe the two phenomena are related?

Let

$$S = \{n \geq 1 : n = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z}\}.$$

We saw $1, 2 \in S$ and $m^2 \in S$ for every integer $m \geq 1$.

Defn (Exponent) Let $n \geq 1$ be an integer and p a prime.

The exponent of p in n, denoted by $e_p(n)$, is the greatest integer e s.t. $p^e | n$.

Eg. $n = 1500 = 2^2 \cdot 3^1 \cdot 5^3$.

$e_2(n) = 2$, $e_3(n) = 1$, $e_5(n) = 3$,

$e_p(n) = 0$, $p > 5$.

Thm. Given an integer $n \geq 1$, $n \in S$ if and only if for every $p | n$ with $p \equiv 3 \pmod 4$, $e_p(n)$ is even.

Lem. Assume $n \in S$ can be written as

$$n = x^2 + y^2, \quad x, y \in \mathbb{Z}.$$

If $p | n$ and $p \equiv 3 \pmod 4$, then $p | x$ and $p | y$.

Consequently, $p^2 | n$.

Pf. If $p|x$, then $p|x^2$. Since $p|n$, $p|y^2 = n - x^2$.

By Euclid, $p|y$. Moreover, $p^2 | x^2$ and $p^2 | y^2$. So

$p^2 | n = x^2 + y^2$. The same argument works for the case $p|y$.

Assume $p \nmid x$, $p \nmid y$. Look at

$$n = x^2 + y^2 \equiv 0 \pmod{p} \qquad (*).$$

There is $x' \in \mathbb{Z}$ s.t. $xx' \equiv 1 \pmod{p}$, which is possible

because of $(p, x) = 1$. Multiplying $(*)$ by x'^2:

$$\underbrace{(xx')^2}_{\equiv 1} + (\underbrace{yx'}_{z})^2 \equiv 0 \pmod{p}$$

$$\Rightarrow 1 + z^2 \equiv 0 \pmod{p} \Rightarrow z^2 \equiv -1 \pmod{p}.$$

So $-1 \, R \, p$. But this is impossible, as $p \equiv 3 \pmod 4$. $\square$

Pf of the necessity part: Let $n \in S$ with

$$n = x^2 + y^2 \quad (**)$$

be minimal s.t. n has a prime factor $p \equiv 3 \pmod 4$

with $e_p(n)$ odd. By Lem, $p \mid x$, $p \mid y$ and $p^2 \mid n$.

Dividing $(**)$ by p^2 gives

$$\underbrace{\frac{n}{p^2}}_{\in \mathbb{Z}} = \Big(\underbrace{\frac{x}{p}}_{\in \mathbb{Z}}\Big)^2 + \Big(\underbrace{\frac{y}{p}}_{\in \mathbb{Z}}\Big)^2 \in S,$$

where $\frac{n}{p^2} < n$ and $e_p\left(\frac{n}{p^2}\right) = e_p(n) - 2$ is odd. A

Contradiction. $\square$

Prop. S is multiplicavely closed: $m, n \in S \Rightarrow mn \in S$.

Pf. If $m = a^2+b^2$ and $n = c^2+d^2$, then

$$mn = \underbrace{(a^2+b^2)}_{|a+bi|^2} \underbrace{(c^2+d^2)}_{|c+di|^2} = \underbrace{(ac-bd)^2+(bc+ad)^2}_{|\underbrace{(a+bi)(c+di)}_{(ac-bd)+(bc+ad)i}|^2} \in S.$$

$\square$

Thm (Fermat-Euler) Every prime $p \equiv 1 \pmod 4$ can be written as a sum of two squares.

Pf. Consider

$$x - ly, \quad x, y \in \{0, 1, \dots, \lfloor \sqrt{p} \rfloor\},$$

where we choose l later. There are

$$(\underbrace{\lfloor\sqrt{p}\rfloor}_{>\sqrt{p}-1}+1)^2 > (\sqrt{p})^2 = p$$

such pairs. Since there are only p residue classes mod p,

there exist $(x_1, y_1) \neq (x_2, y_2)$ s.t.

$$x_1 - \ell y_1 \equiv x_2 - \ell y_2 \pmod{p}$$

$$\Rightarrow \underbrace{x_1 - x_2}_{x} \equiv \ell \underbrace{(y_1 - y_2)}_{y} \pmod{p} \quad (***).$$

At least one of x, y is nonzero, say x. Also,

$0 < |x| \leq \lfloor\sqrt{p}\rfloor < \sqrt{p}$ and $0 \leq |y| \leq \lfloor\sqrt{p}\rfloor < \sqrt{p}$.

Since $-1\ R\ p$, we can find an $\ell \in \mathbb{Z}$ s.t. $\ell^2 \equiv -1 \pmod{p}$.

Squaring $(***)$ leads to $x^2 \equiv \ell^2 y^2 \equiv -y^2 \pmod{p}$.

So $x^2 + y^2 \equiv 0 \pmod{p}$.

Thus $x^2 + y^2$ is a multiple of p. But

$$0 < x^2 + y^2 < (\sqrt{p})^2 + (\sqrt{p})^2 = 2p.$$

Hence $p = x^2 + y^2$. $\square$